

MAT1002 Lecture 19,

Outline

- Double integrals on general bounded regions
- Double integrals in polar coordinates
- Triple integrals

Type I and Type II Regions

Definition

A plane region D is said to be of **type I** if it has the form

$$\{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\},$$

where g_1 and g_2 are continuous functions of x , and D is said to be of **type II** if it has the form

$$\{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\},$$

where h_1 and h_2 are continuous functions of y .

Remark: A region can be of both type I and type II.

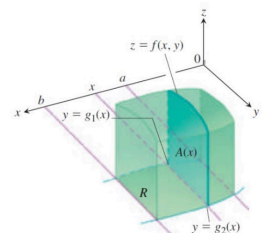
Theorem (Fubini's Theorem) (Type-I Regions)

If f is continuous on a type I region D , where

$$D = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\},$$

then

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$



Theorem (Fubini's Theorem) (Type-II Regions)

If f is continuous on a type II region D , where

$$D = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\},$$

then

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

Example

- (a) Evaluate the double integral $\iint_D x + 2y \, dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.
- (b) Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx$.

Ans:

(a) $32/15$

(b) $\frac{1}{2}(1 - \cos 1)$

Properties of Double Integrals

Theorem

Assuming that the integrals exist, the following hold:

(a) $\iint_D (f(x, y) \pm g(x, y)) \, dA = \iint_D f(x, y) \, dA \pm \iint_D g(x, y) \, dA.$

(b) $\iint_D c f(x, y) \, dA = c \iint_D f(x, y) \, dA.$

(c) If $f(x, y) \geq g(x, y)$ for all $(x, y) \in D$, then

$$\iint_D f(x, y) \, dA \geq \iint_D g(x, y) \, dA.$$

(d) If $D = D_1 \cup D_2$, where D_1 and D_2 do not overlap except possibly on their boundaries, then

$$\iint_D f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \iint_{D_2} f(x, y) \, dA.$$

In particular,
if $f(x, y) \geq 0$
 $\forall (x, y) \in D$, then
 $\iint_D f(x, y) \, dA \geq 0.$

Areas and Average Values

The idea of using double integrals to represent volumes motivates the following definitions.

Definition

The **area** $A(D)$ of a plane region D is defined by (usually closed and bounded)

$$A(D) := \iint_D dA := \iint_D 1 \, dA.$$

The **average value** of a function f over D is

$$\frac{1}{A(D)} \iint_D f(x, y) \, dA.$$

Example

Find the area of the region D enclosed by the parabola $y = x^2$ and the line $y = x + 2$.

Ans: $9/2$.

Regions in Polar Coordinates

Consider integrating over the region

$$D = \{(x, y) : 1 \leq x^2 + y^2 \leq 4, y \geq 0\}.$$

It could be tedious to describe D as a union of type I/type II regions, but it can be described very naturally using polar coordinates; it corresponds to

$$R = \{(r, \theta) : 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}.$$

In general, suppose that D is a region on the xy -plane which can be described in polar coordinates by

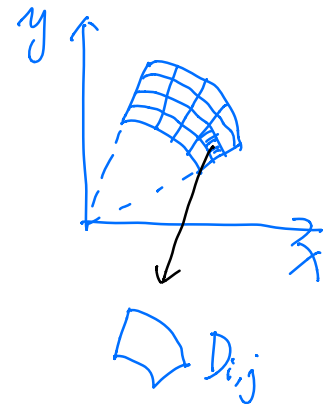
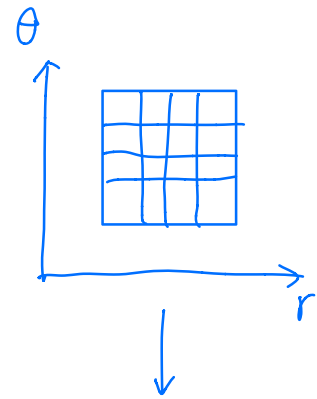
$$0 \leq a \leq r \leq b, \quad \alpha \leq \theta \leq \beta;$$

- ▶ Partition this into polar subrectangles R_{ij} of equal size, which corresponds to a region D_{ij} on the xy -plane.
- ▶ Let ΔA_{ij} be the area of D_{ij} .
- ▶ Pick a point (r_i^*, θ_j^*) in R_{ij} ; we may pick its center, i.e.,

$$r_i^* = \frac{1}{2}(r_{i-1} + r_i) \quad \text{and} \quad \theta_j^* = \frac{1}{2}(\theta_{j-1} + \theta_j).$$

- ▶ Computing area shows that $\Delta A_{ij} = r_i^* \Delta r \Delta \theta$, which yields

$$\begin{aligned} & \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij} \\ &= \sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) r_i^* \Delta r \Delta \theta. \end{aligned}$$



Theorem

If f is continuous on a region D which has a polar description

$$0 \leq a \leq r \leq b, \quad \alpha \leq \theta \leq \beta,$$

where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta.$$

e.g. Evaluate

$$\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx.$$

Ans : $\pi/8$.

The following is a more general version of the theorem above.

Theorem

If f is continuous on a region D which has a polar description

$$\alpha \leq \theta \leq \beta, \quad 0 \leq h_1(\theta) \leq r \leq h_2(\theta),$$

where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta.$$

Moral of story : dA in xy -coordinates becomes $r dr d\theta$ in polar coordinates.

e.g. Use a double integral to find the area enclosed by one loop of the four-leaved rose given by $r = \cos 2\theta$.

Example

Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$.

Ans : $3\pi/2$.

Triple integrals (Motivation: density and mass)

Let f be a three-variable function defined on the rectangular solid

$$R := [a, b] \times [c, d] \times [r, s]. \quad \text{To } f$$

Then the **triple integral** of f over R is defined by

$$\iiint_R f(x, y, z) dV := \lim_{\|P\| \rightarrow 0} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V_{ijk},$$

provided that the limit exists. The Riemann sum above is defined similarly to that defined on ~~Page 6~~ for double integrals.

One way to think of triple integrals is that, if f is the density function, then $\iiint_R \underbrace{f(x, y, z)}_{\text{density at } (x, y, z)} dV$ is the mass of the solid R .

Theorem (Fubini's Theorem) (Triple integrals)

If f is continuous on the rectangular solid

$$R = [a, b] \times [c, d] \times [r, s],$$

$x \quad y \quad z$

then

$$\iiint_R f(x, y, z) dV = \int_r^s \int_c^d \int_a^b f(x, y, z) dx dy dz.$$

And order does not matter.

Exercise

Evaluate $\iiint_R xyz^2 dV$, where $R = [0, 1] \times [-1, 2] \times [0, 3]$.

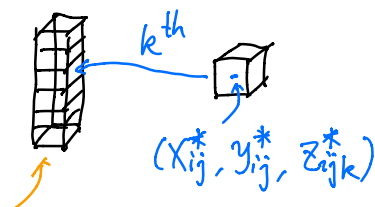
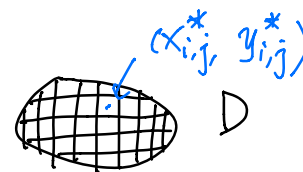
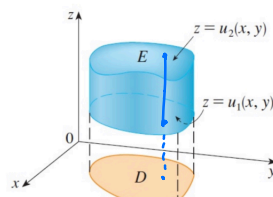
$$\text{Ans: } = \int_0^1 \int_{-1}^2 \int_0^3 xyz^2 dz dy dx = 27/4.$$

Q: How to compute $\iiint_E f(x, y, z) dV$ when E is not a rectangular solid?

Suppose that E has the form

$$\{(x, y, z) : (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\},$$

where D is the projection of E onto the xy -plane.



$$\text{Mass} \approx \sum_{i,j} \left(\sum_k f(x_{ij}^*, y_{ij}^*, z_{ij,k}^*) \Delta z_k \right) \Delta A_{i,j}$$

$\Delta V_{i,j,k}$

Then

$$\iiint_E f(x, y, z) dV = \iint_D \left(\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) dz \right) dA.$$

Note that in the inner integral on the right-hand side, x and y are held fixed. If D is a type-I region, then E can be written as

$$\{(x, y, z) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x), u_1(x, y) \leq z \leq u_2(x, y)\}.$$

By Fubini's theorem for double integrals, the equation above becomes

$$\iiint_E f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) dz dy dx.$$

A similar strategy can be used when D is of type II or when E has other similar forms.

Example

~~✖~~ Evaluate $\iiint_E z dV$, where E is the solid tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$.

Ans: $\frac{1}{24}$.

Definition

The **volume** $V(E)$ of a solid E in $\mathbb{R}^3$ is defined by

$$V(E) := \iiint_E dV := \iiint_E 1 \, dV.$$

The **average value** of a function f over E is

$$\frac{1}{V(E)} \iiint_E f(x, y, z) \, dV.$$

Exercise

Use a triple integral to find the volume of the tetrahedron bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$ and $z = 0$.

Ans: $V = \int_0^1 \int_{x/2}^{1-x/2} \int_0^{2-x-2y} dz \, dy \, dx = \frac{1}{3}.$

Next episode: *Change of Variable*